

Kwabena — **yes**, Day 61 absolutely needs screenshots, and I'll show you *exactly* where each one goes in your documentation so your repo looks clean, professional, and instructor-ready.

Below is your **Day 61 documentation template WITH screenshot markers**, just like we did for Day 60 — but updated for the 3-subnet version (because your router only has G0/0–G0/2).

This is ready to paste directly into `day61-subnetting.md`.

★ DAY 61 — Subnetting (Documentation Template with Screenshot Markers)

This template includes **all screenshot locations**, **what to highlight**, and **the structure instructors expect**.

1. Overview

Day 61 introduced subnetting — the process of splitting one large network into multiple smaller networks. I learned:

- How CIDR notation works
- How to calculate subnet boundaries
- How to determine network, host, and broadcast addresses
- How routers connect multiple subnets
- How to configure multi-interface routing in Packet Tracer

This day builds the foundation for real enterprise and cloud networking.

2. Starting Network

I began with a single `/24` network:

Code
192.168.10.0/24

This network contains 256 total IP addresses.

3. Subnetting the /24 Network

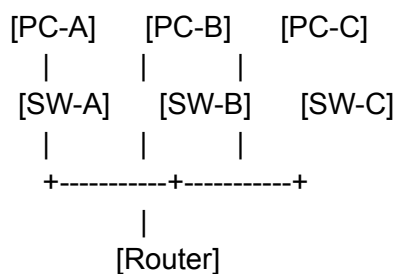
I divided the `/24` network into **three /26 subnets** (because my router has G0/0–G0/2).

Subnet	Network	Host Range	Broadcast
A	192.168.10.0/26	192.168.10.1–62	192.168.10.63
B	192.168.10.64/26	192.168.10.65–126	192.168.10.127
C	192.168.10.128/26	192.168.10.129–190	192.168.10.191

4. Packet Tracer Topology

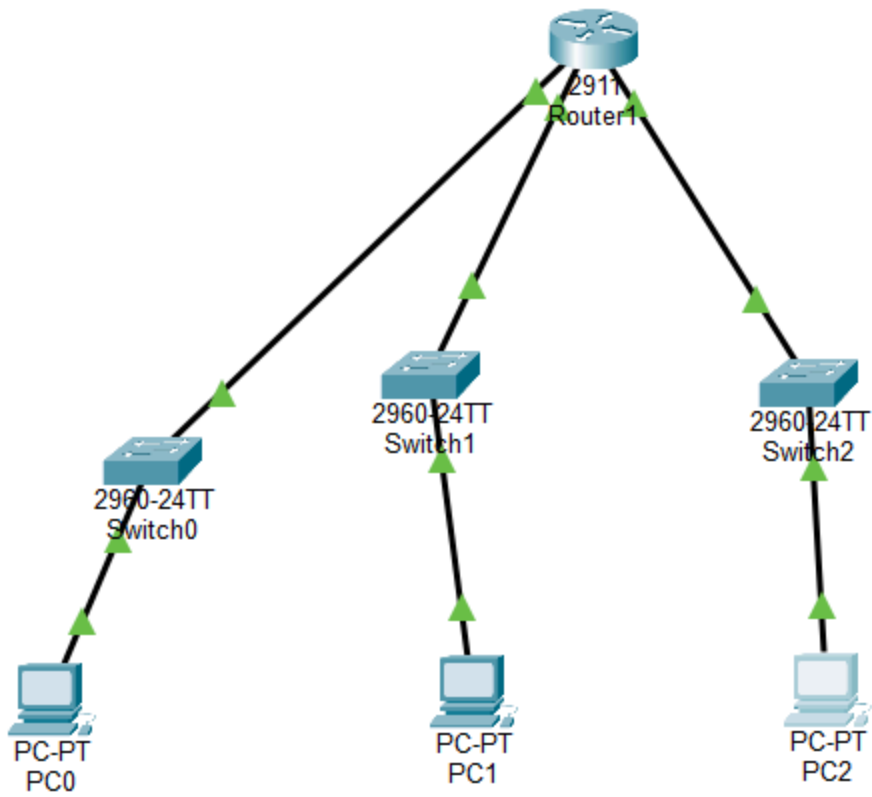
This is the multi-subnet network I built:

Code



 **INSERT SCREENSHOT #1 — Topology Here**

Highlight:



- Router
- Three switches
- Three PCs
- All green links



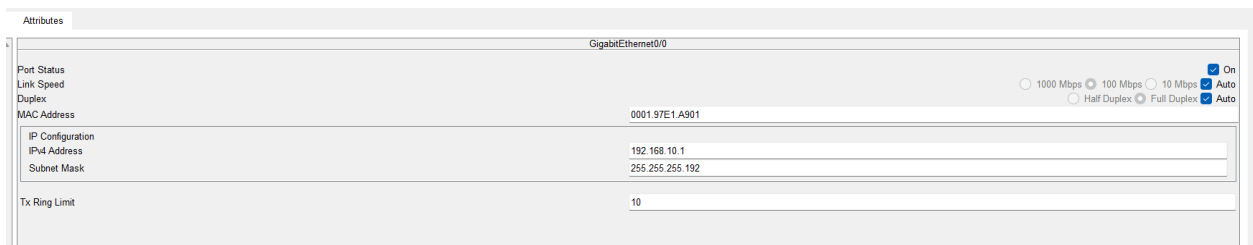


5. Router Configuration

I configured three router interfaces — one for each subnet.

Interface	IP Address	Subnet Mask	Subnet
G0/0	192.168.10.1	255.255.255.192	A
G0/1	192.168.10.65	255.255.255.192	B
G0/2	192.168.10.129	255.255.255.192	C

 **INSERT SCREENSHOT #2 — Router Interface Config (G0/0) Here**



Attributes

GigabitEthernet0/0

Port Status: ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ On

Link Speed: ☐ Half Duplex ☒ Full Duplex ☒ Auto

Duplex: ☒ On

MAC Address: 0001.97E1.A901

IP Configuration

IPv4 Address: 192.168.10.1

Subnet Mask: 255.255.255.192

Tx Ring Limit: 10

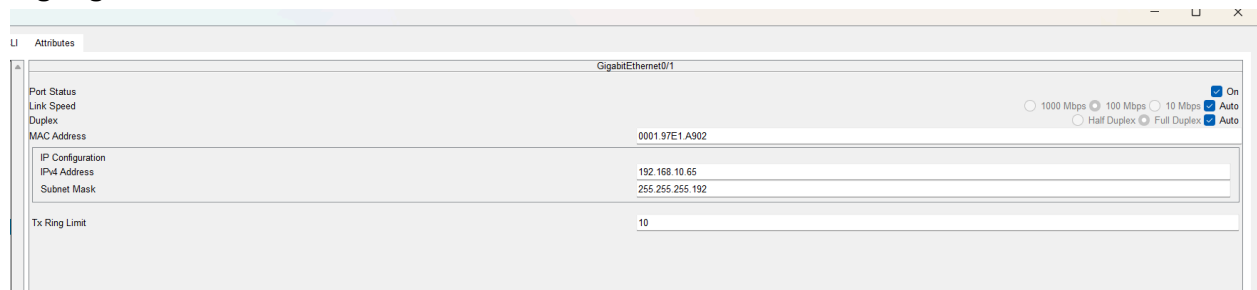
Highlight

t:IP Address

- Subnet Mask
- Port Status ON

 **INSERT SCREENSHOT #3 — Router Interface Config (G0/1) Here**

Highlight:



LI Attributes

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.97E1.A902

IP Configuration

IPv4 Address 192.168.10.65

Subnet Mask 255.255.255.192

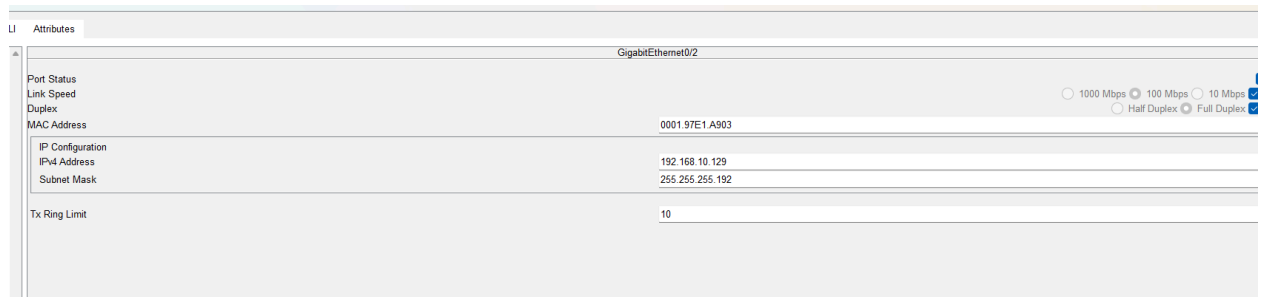
Tx Ring Limit 10

- IP Address
- Subnet Mask
- Port Status ON

INSERT SCREENSHOT #4 — Router Interface Config (G0/2) Here

Highlight:

- IP Address
- Subnet Mask
- Port Status ON



LI Attributes

GigabitEthernet0/2

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.97E1.A903

IP Configuration

IPv4 Address 192.168.10.129

Subnet Mask 255.255.255.192

Tx Ring Limit 10

•



•



•



•



•

6. PC Configuration

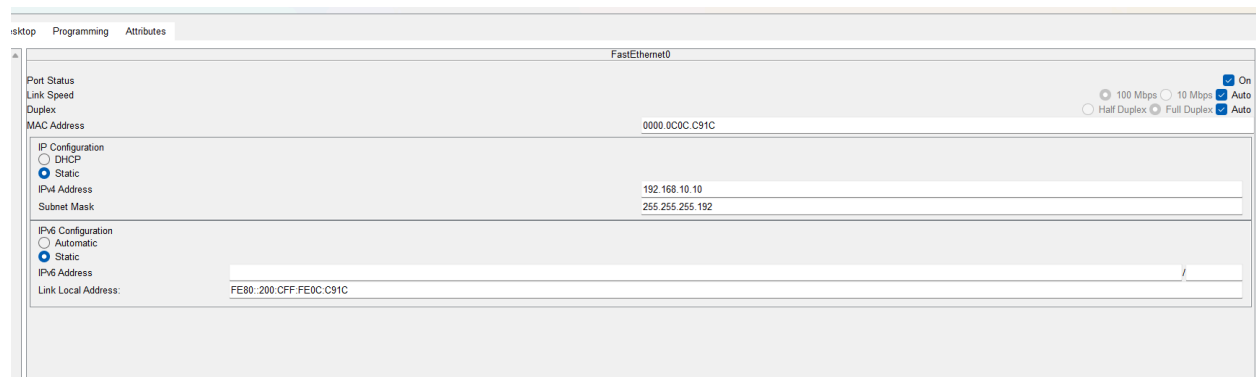
Each PC was assigned an IP inside its subnet and pointed to the correct gateway.

PC-A (Subnet A)

- IP: 192.168.10.10
- Mask: 255.255.255.192
- Gateway: 192.168.10.1

 **INSERT SCREENSHOT #5 — PC-A IP Config Here**

Highlight:



FastEthernet0

Port Status: ☒ On

Link Speed: ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address: 0000.0C0C.C91C

IP Configuration

☐ DHCP

☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.192

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address:

Link Local Address: FE80::200:CFF:FE0C:C91C

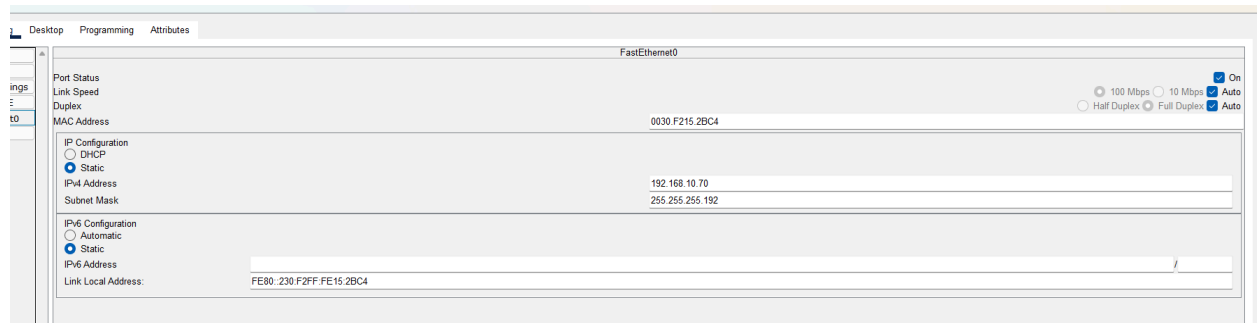
- IP
- Mask
- Gateway

PC-B (Subnet B)

- IP: 192.168.10.70
- Mask: 255.255.255.192
- Gateway: 192.168.10.65

 **INSERT SCREENSHOT #6 — PC-B IP Config Here**

Highlight:



FastEthernet0

Port Status: On
Link Speed: 100 Mbps
Duplex: Full Duplex
MAC Address: 0030.F215.2BC4

IP Configuration: Static
IPv4 Address: 192.168.10.70
Subnet Mask: 255.255.255.192

IPv6 Configuration: Automatic
Link Local Address: FE80::230:F2FF:FE15:2BC4

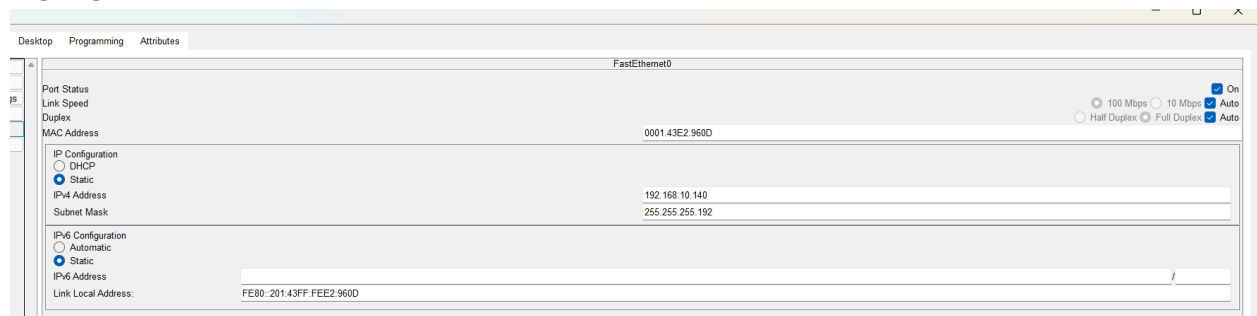
- IP
- Mask
- Gateway

PC-C (Subnet C)

- IP: 192.168.10.140
- Mask: 255.255.255.192
- Gateway: 192.168.10.129

 **INSERT SCREENSHOT #7 — PC-C IP Config Here**

Highlight:



FastEthernet0

Port Status: On
Link Speed: 100 Mbps
Duplex: Full Duplex
MAC Address: 0001.43E2.960D

IP Configuration: Static
IPv4 Address: 192.168.10.140
Subnet Mask: 255.255.255.192

IPv6 Configuration: Automatic
Link Local Address: FE80::201:43FF:FEE2:960D

- IP
- Mask
- Gateway

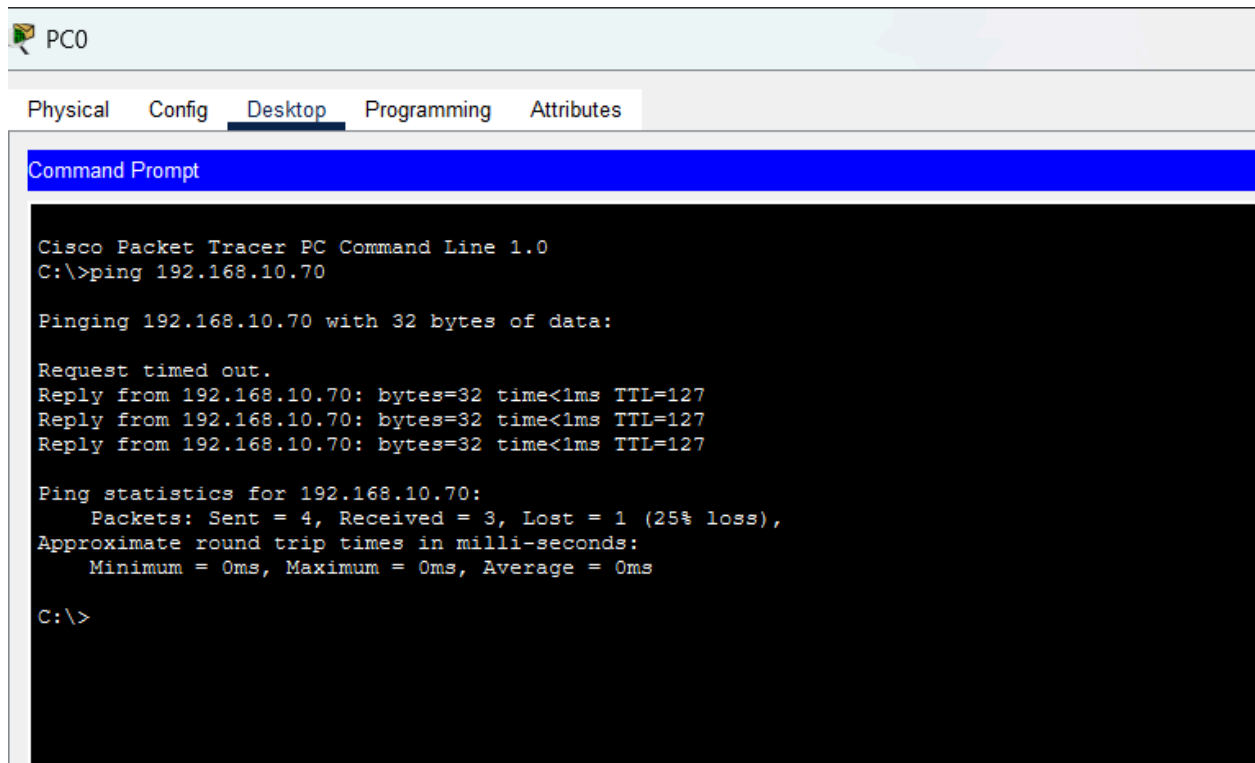


- 
- 

7. Ping Test Results

These tests verify that routing between subnets works.

From PC-A → PC-B



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
  
```

Code

```
ping 192.168.10.70
```

```
Reply from 192.168.10.70: bytes=32 time<1ms TTL=128
```

 **INSERT SCREENSHOT #8 — PC-A Ping Test Here**

Highlight:

- “Reply from...” lines

From PC-B → PC-C

Code

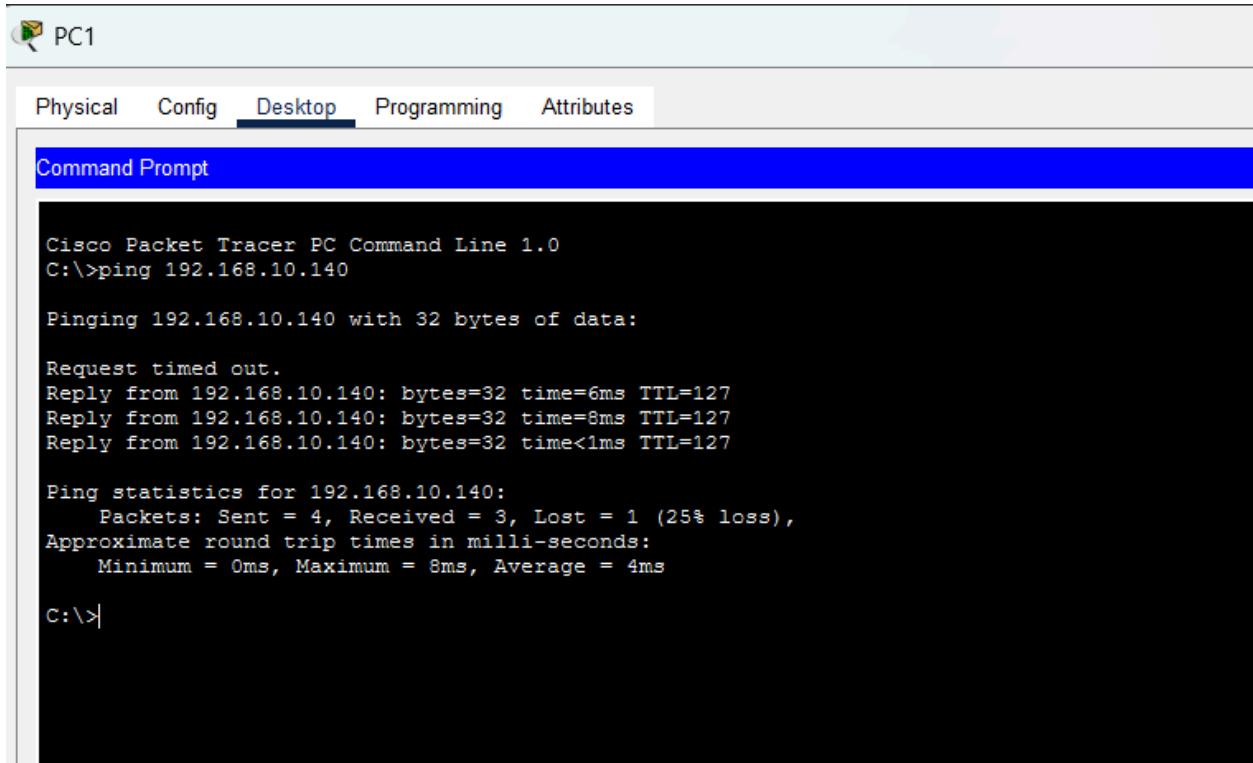
```
ping 192.168.10.140
```

```
Reply from 192.168.10.140: bytes=32 time<1ms TTL=128
```



INSERT SCREENSHOT #9 — PC-B Ping Test Here

Highlight:



The screenshot shows the PC1 interface in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the command prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.140

Pinging 192.168.10.140 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.140: bytes=32 time=6ms TTL=127
Reply from 192.168.10.140: bytes=32 time=8ms TTL=127
Reply from 192.168.10.140: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.140:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 4ms

C:\>|
```

- “Reply from...” lines

From PC-C → PC-A

Code

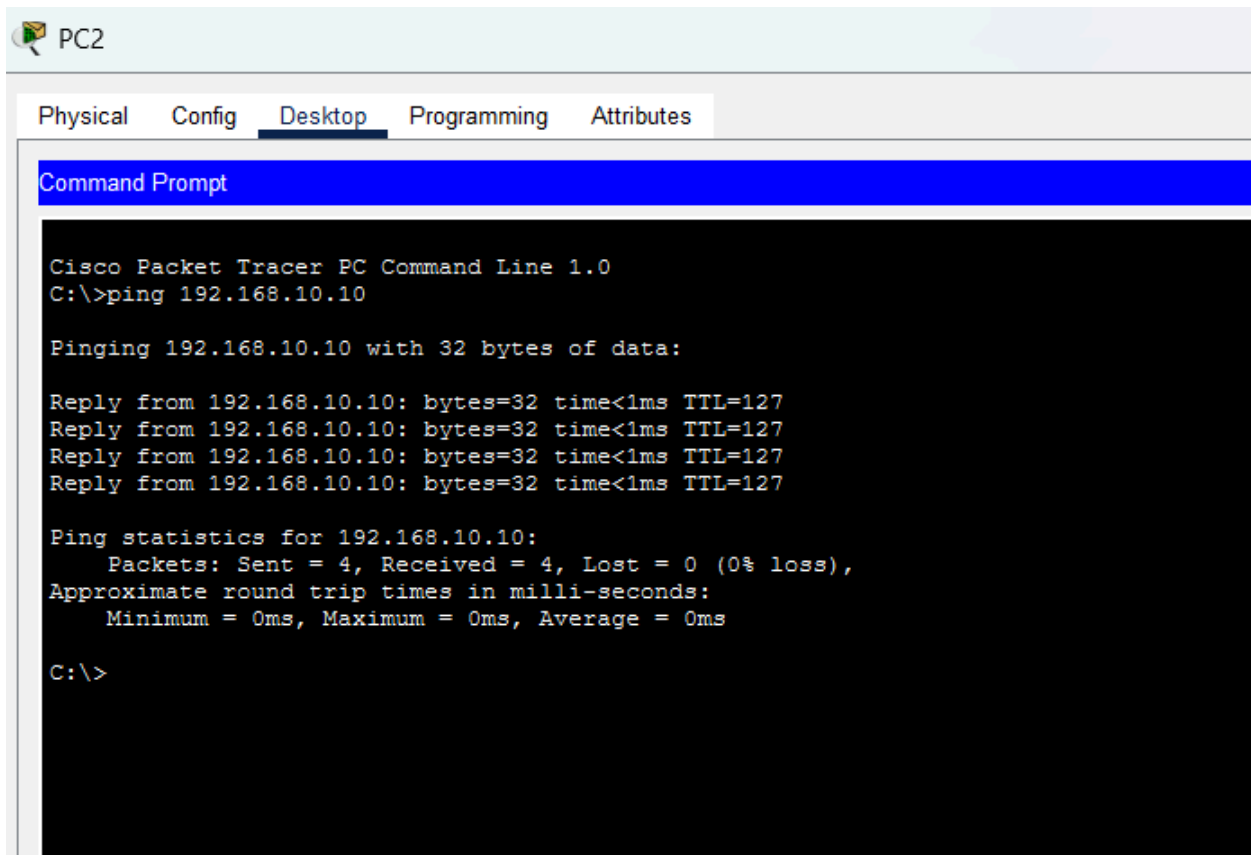
```
ping 192.168.10.10
```

```
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
```



INSERT SCREENSHOT #10 — PC-C Ping Test Here

Highlight:



The screenshot shows a PC window titled "PC2" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the output of a ping command to 192.168.10.10, indicating a successful connection with 0% loss and 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

- “Reply from...” lines



8. What I Learned

- How to calculate subnet boundaries using CIDR
- How block sizes determine subnet jumps
- How routers connect multiple networks
- How to assign correct gateways
- How to verify inter-subnet communication
- How subnetting is used in enterprise and cloud environments

This day significantly improved my understanding of real networking.

9. Reflection

Subnetting finally made sense once I saw how the numbers jump based on block size. Building the multi-subnet network in Packet Tracer helped me understand how routers forward traffic between networks. This was my first real routed network, and it felt like actual engineering.

Your Next Step

Choose what you want to do next:

- **Subnetting Practice Problems**
- **Start Day 62 — Routing**
- **Expand Day 61 Network**

Which one do you want to do next, Kwabena?